

Exercise Sheet Solutions

Question 1

The table below lists three different estimators of the Bernoulli mean parameter, where n is the sample size, $S = \sum_{i=1}^n X_i$ is the sum of Bernoulli samples. In deriving the MAP and the Bayesian estimators, a $Beta(\alpha, \beta)$ prior distribution was used. (For derivation details see the Tutorial slides from week 2).

Method	Estimator
Maximum Likelihood	$\hat{\theta}_{ML} = S/n$
Maximum A Posteriori	$\hat{\theta}_{MAP} = \frac{S+\alpha-1}{n+\alpha+\beta-2}$
Bayesian posterior mean	$\hat{\theta}_{Bayes} = \frac{S+\alpha}{n+\alpha+\beta}$

1. Determine the bias and variance of these estimators.
2. How does the choice of prior influence the bias and variance of the MAP and Bayesian estimators? Which choices of prior parameters α, β reduce the variance of the estimators, and what happens to the bias with those choices. To answer this question, try at least 3 different values for α, β .
3. Compare the bias and variance of the Bayesian estimator to those of the ML and MAP, and comment on the potential implications of your findings for practical applications.

Answer for Question 1

1. The question asks to do an analysis of the three estimators in terms of the frequentist bias and variance. At the lecture, we already determined the bias and variance of the first 2 of the estimators, and we obtained:

$$\text{Bias}(\hat{\theta}_{MLE}) = 0; \quad \text{Var}(\hat{\theta}_{MLE}) = \frac{\theta(1-\theta)}{n}$$

$$\text{Bias}(\hat{\theta}_{MAP}) = E[\hat{\theta}_{MAP}] - \theta = \frac{n\theta + \alpha - 1}{n + \alpha + \beta - 2} - \theta; \quad \text{Var}(\hat{\theta}_{MAP}) = \frac{n\theta(1-\theta)}{(n + \alpha + \beta - 2)^2}$$

Now we compute the Bias of Bayesian Mean estimator. Taking the expectation,

$$E[\hat{\theta}_{\text{Bayes}}] = E\left[\frac{\alpha + S}{\alpha + \beta + n}\right] = \frac{\alpha + n\theta}{\alpha + \beta + n},$$

since $E[S] = n\theta$. Hence, the bias is:

$$\text{Bias}(\hat{\theta}_{\text{Bayes}}) = E[\hat{\theta}_{\text{Bayes}}] - \theta = \frac{\alpha + n\theta}{\alpha + \beta + n} - \theta.$$

Finally, the variance of the Bayesian estimator is:

$$\text{Var}(\hat{\theta}_{\text{Bayes}}) = \frac{1}{(n + \alpha + \beta)^2} \cdot \text{Var}\left(\sum_{i=1}^n X_i\right) = \frac{n\theta(1 - \theta)}{(n + \alpha + \beta)^2}.$$

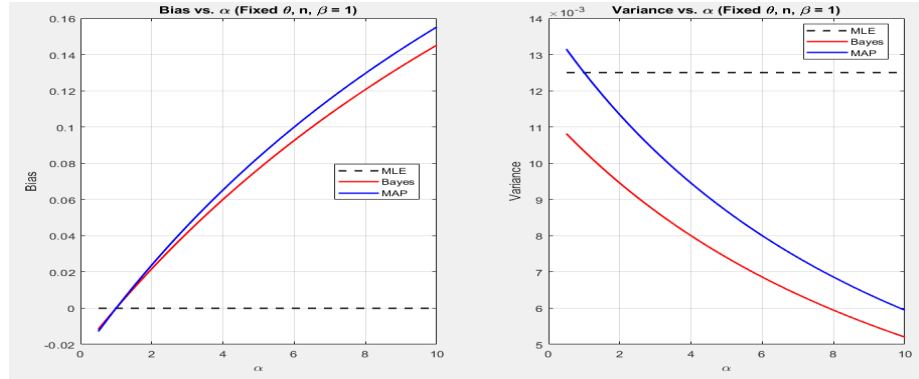
Summary:

Estimator	Bias	Variance
MLE	0 (Unbiased)	$\frac{\theta(1-\theta)}{n}$
MAP	$\frac{\alpha+n\theta-1}{\alpha+\beta+n-2} - \theta$	$\frac{n\theta(1-\theta)}{(n+\alpha+\beta-2)^2}$
Bayesian Mean	$\frac{\alpha+n\theta}{\alpha+\beta+n} - \theta$	$\frac{n\theta(1-\theta)}{(n+\alpha+\beta)^2}$

2. The larger the α and β – meaning the more informative i.e. different from uniform the Beta prior – the smaller the variance. And the larger the bias.

3. When $\alpha + \beta > 2$, the variance of $\hat{\theta}_{\text{MAP}}$ is less than the variance of $\hat{\theta}_{\text{MLE}}$. The variance of $\hat{\theta}_{\text{Bayes}}$ is always smaller since $\alpha, \beta > 0$.

The figure below shows how the bias and variance of the 3 estimators varies with α , while keeping $\theta = 0.5, n = 20$ and $\beta = 1$ fixed.



Question 2

- Derive the MLE and MAP estimators for the parameter (λ) of a Poisson distribution (hint: you've done the MLE for already in a previous exercise sheet). For the latter, use a Gamma distribution as the prior.
- Analyse and compare the bias and variance of the MLE and MAP estimators obtained.

Answer for Question 2

Given independent and identically distributed (i.i.d.) random variables:

$$X_i \sim \text{Poisson}(\lambda), \quad i = 1, 2, \dots, n,$$

the probability mass function (PMF) is (you can look this up on wiki, no need to memorise it):

$$P(X_i = x_i | \lambda) = \frac{\lambda^{x_i} e^{-\lambda}}{x_i!}, \quad x_i \in \mathbb{N}_0.$$

Maximum Likelihood Estimator (MLE):

The likelihood function for a sample $X_1, X_2, \dots, X_n$ is:

$$L(\lambda | X_1, \dots, X_n) = \prod_{i=1}^n \frac{\lambda^{X_i} e^{-\lambda}}{X_i!}.$$

Taking the logarithm:

$$\log L(\lambda | X_1, \dots, X_n) = \sum_{i=1}^n X_i \log \lambda - n\lambda - \sum_{i=1}^n \log X_i!.$$

Differentiating with respect to λ and setting to zero we obtain the ML estimator (we knew this from a previous homework):

$$\sum_{i=1}^n \frac{X_i}{\lambda} = n \quad \Rightarrow \quad \hat{\lambda}_{\text{MLE}} = \frac{1}{n} \sum_{i=1}^n X_i.$$

Now compute the MAP estimator, using a Gamma prior (you can find the PDF of the Gamma distribution on wiki, no need to memorise it):

$$\lambda \sim \text{Gamma}(\alpha, \beta),$$

where the PDF of the Gamma distribution is:

$$p(\lambda) = \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda}, \quad \lambda > 0.$$

The posterior distribution is proportional to:

$$p(\lambda | X_1, \dots, X_n) \propto L(\lambda) p(\lambda).$$

Expanding both terms:

$$p(\lambda|X_1, \dots, X_n) \propto \lambda^{\sum_{i=1}^n X_i} e^{-n\lambda} \cdot \lambda^{\alpha-1} e^{-\beta\lambda}.$$

Rewriting:

$$p(\lambda|X_1, \dots, X_n) \propto \lambda^{\alpha+S-1} e^{-(\beta+n)\lambda},$$

where $S = \sum_{i=1}^n X_i$. To find the MAP estimator, we maximize the posterior:

$$\log p(\lambda|X_1, \dots, X_n) = (\alpha + S - 1) \log \lambda - (\beta + n)\lambda.$$

Differentiating with respect to λ :

$$\frac{d}{d\lambda} [(\alpha + S - 1) \log \lambda - (\beta + n)\lambda] = \frac{\alpha + S - 1}{\lambda} - (\beta + n) = 0.$$

Solving for λ :

$$\hat{\lambda}_{MAP} = \frac{\alpha + S - 1}{\beta + n}.$$

Frequentist analysis of Bias and Variance

Since $X_i \sim \text{Poisson}(\lambda)$, we have:

$$\mathbb{E}[X_i] = \lambda, \quad \text{Var}(X_i) = \lambda.$$

Thus, for $\hat{\lambda}_{MLE}$:

$$\mathbb{E}[\hat{\lambda}_{MLE}] = \lambda \quad \Rightarrow \quad \text{Bias}_{\text{freq}}(\hat{\lambda}_{MLE}) = 0.$$

The variance is:

$$\text{Var}_{\text{freq}}(\hat{\lambda}_{MLE}) = \frac{\lambda}{n}.$$

Next, we determine the Bias and Variance of the MAP estimator. Denoting again $S = \sum_{i=1}^n X_i$ and taking expectation:

$$\mathbb{E}[\hat{\lambda}_{MAP}] = \frac{\alpha + \mathbb{E}[S] - 1}{\beta + n} = \frac{\alpha + n\lambda - 1}{\beta + n}.$$

So the frequentist bias is:

$$\text{Bias}(\hat{\lambda}_{MAP}) = \frac{\alpha + n\lambda - 1}{\beta + n} - \lambda = \lambda \left(\frac{n}{\beta + n} - 1 \right) + \frac{\alpha - 1}{\beta + n}.$$

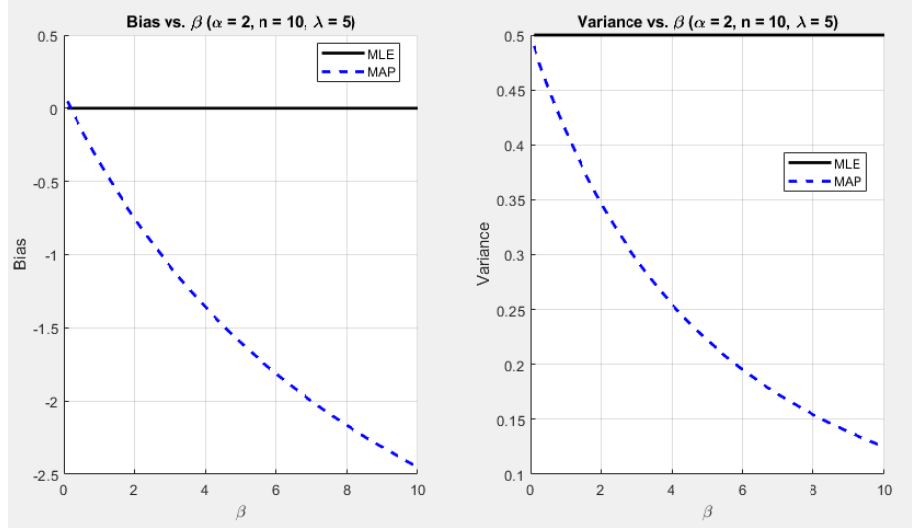
The variance is:

$$\text{Var}(\hat{\lambda}_{MAP}) = \frac{n\lambda}{(\beta + n)^2}.$$

The following table summarises the results.

2. We see that the variance of the MAP estimator is always less than that of the MLE in this case. To further compare, the plots below illustrate how the bias and variance change with the choice of prior parameter β . Note, that negative bias is just as bad as positive bias.

Estimator	Bias	Variance
MLE	0	$\frac{\lambda}{n}$
MAP	$\lambda \left(\frac{n}{\beta+n} - 1 \right) + \frac{\alpha-1}{\beta+n}$	$\frac{n\lambda}{(\beta+n)^2}$



Question 3

Suppose we have a data set for a medical diagnosis problem, and we want to use decision trees.

1. What could go wrong if your tree is as deep as it takes to classify all training points correctly?
2. How would you use confidence bounds to select an appropriate depth for the tree? (Which bounds would you use, why, and how? how would you decide this?)

Answer for Question 3

1. The decision tree classifier can easily overfit if it is grown too large. That is, even if it classifies all training points correctly it does not mean that it will classify previously unseen points correctly.

2. If there is enough labelled data to afford an independent test set, we can use the test set bound as a criterion to optimise when selecting the depth of the tree. This is a tight bound on the true generalisation error, and ensures with high confidence that the observed error rate as measured on the test set is not just by chance. Such confidence guarantees are more important in safety-critical

applications, like medical, or autonomous driving applications, where we don't want to take chances unnecessarily.

If there is not enough labelled data, then we can use the Occam's razor bound computed on the training set instead of the test set bound. This is typically less tight, unless we have strong prior knowledge about the application problem as to which classifiers are more likely to be returned by the training algorithm.

Question 4

a) Suppose you trained a classifier on some data set and it achieves zero error on the training set.

- (i) Is the 0 test error necessarily a good thing? Why or why not?
- (ii) Can you use the Occam's razor bound to argue that your classifier is good? Why or why not?
- (iii) On an independent test set (which wasn't used for training) of size 30, you get 4 errors. Give a 90% and a 95% confidence upper bounds on the true generalisation error of your classifier. State what method you used to obtain it and justify why the method use is appropriate.

b) A machine learning model is trained to classify emails as spam or not spam. To evaluate its accuracy, we test it on a random sample of 1000 emails and find that it correctly classifies 850 of them.

- (i) Using the corollary of the test set bound, construct a 95% confidence upper bound for the true classification accuracy of the model.
- (ii) Interpret the notion of confidence interval in the context of email classification.

Answer for Question 4

a) (i) A 0 test error is a good thing, but it can still be misleading if the test set happened to be too small or not sufficiently representative. The test set bound counters this by giving an upper bound at any specified confidence level, which takes into account the size of the test set.

(ii) Yes, with some limitations. The bound provides a way to argue that a classifier generalizes well if it both (1) fits the training data well (low empirical error) and (2) has a low description length or self-information relative to the distribution of our prior bets. However, it does not prove that a classifier is good, only that its generalization error is bounded under certain conditions. The conditions are: The sample is i.i.d., and the prior bets were made before seeing the sample.

If the bound is small then the true error must be small with high confidence (where confidence means high proportion of training sets to which the bound applies). If the bound is large, it doesn't necessarily mean the classifier is bad.

(iii) We can use the corollary on the test set bounds. To give the 90% confidence upper bound, set $\delta = 0.1$. Applying the bound

$$c_{\mathcal{D}} \leq \hat{c}_S + \sqrt{\frac{\ln \frac{1}{\delta}}{2n}}$$

Substitute values:

$$c_{\mathcal{D}} \leq 4/30 + \sqrt{\frac{\ln(1/0.1)}{60}} \approx 0.3292$$

To give the 95% confidence upper bound, set $\delta = 0.05$. Applying the bound

$$c_{\mathcal{D}} \leq 4/30 + \sqrt{\frac{\ln \frac{1}{0.05}}{60}} \approx 0.3568$$

The method is appropriate because we used the test set errors to compute it.

b. (i) Again we use the corollary of the test set set bound,

$$c_{\mathcal{D}} \leq \hat{c}_S + \sqrt{\frac{\ln \frac{1}{\delta}}{2n}}$$

Substitute values:

$$c_{\mathcal{D}} \leq (1000 - 850)/1000 + \sqrt{\frac{\ln 20}{2000}} \approx 0.1887.$$

(ii) The observed test error is 0.15, but this could be just a matter of luck getting easy test points, while the bound tells you that with this sample size you can be 95% confident that the error is no larger than 0.1887.

In addition, a confidence interval provides both an upper bound and a lower bound with high confidence. A 95% confidence interval means that if we repeatedly took different random samples of 1000 emails and computed confidence intervals, then about 95% of those intervals would contain the true accuracy of the model.